



Raja Manickam sharing his ideas at the Indian Electronics Week 2025, in Bengaluru

Globally, a few major players, TSMC, Nvidia, Intel (under restructuring), AMD, Micron, Samsung, among others, dominate the semiconductor industry. Fewer than ten companies effectively control the global market. Their scale and operating structures shape industry behaviour. While there is nothing inherently wrong with this model, it is centralised and resource-intensive.

India need not emulate this structure. We have an opportunity to disrupt it, guided by frugal engineering. This is not merely about spending less; it is about rethinking how we design, manufacture, and distribute semiconductors.

Rather than investing \$10 billion into a mega fab in one city, we could build smaller, decentralised fabs across the country. A fab consumes vast resources and is highly energy-intensive. Built smaller, these fabs could be locally managed, more inclusive, and

tailored to regional needs.

Historically, fabs have focused on logic and memory, optimised for maximum transistor density, as seen with companies like Nvidia. However, there is a growing shift towards subsystems: sensors, MEMS, analogue components, and chiplets. These form the 'beyond compute' ecosystem, complementing rather than replacing the processor. They do not require mega fabs. Distributed manufacturing is

more appropriate.

Earlier, semiconductor development focused heavily on transistor modelling, layout, and scaling. Today, engineers must design for application-specific systems in real-world contexts, particularly in sensing and embedded domains. Understanding the end use is vital. This evolution gives India a timely opportunity.

With a large base of capable engineers and a range of local challenges, India is a natural testbed for innovation. My sister, a doctor trained in India, often says her time here made her a better physician due to the diversity and volume of cases she encountered. Similarly, Indian engineers face issues the world is trying to solve: food security, clean water, climate change. Indian engineering can deliver global solutions rooted in affordability and relevance.

At traditional firms like TI and

National Semiconductor, product development typically began with a business case: What is the ROI, and can customers afford it? The focus was largely on affluent markets. India can adopt a different approach, which includes solving meaningful problems first, and then building affordable technology around them.

Consider a problem approached by Nvidia compared to an Indian startup. Nvidia might develop a high-end chip requiring advanced cooling systems. An Indian startup might offer an 80-90% effective solution at a fraction of the cost—practical and accessible.

Much of the semiconductor industry has been driven by the economics of silicon focussed on shrinking size and reducing cost. However, material science is opening new avenues. Smaller wafer sizes, such as 15.24cm (6-inch) or 20.32cm (8-inch), can still yield excellent outcomes when combined with new materials, especially in power electronics and high-voltage applications.

Opportunities in semiconductors are vast and not only in design but also in device physics, materials science, and system integration. For students in electronics or semiconductor fields, this is a remarkable time. Flexibility and curiosity will be key.

To conclude, I propose a shift towards agile, distributed fabs requiring less than \$10 million in investment as a viable alternative to billion-dollar mega-projects. This model aligns with India's needs and strengths, enabling us to lead with purpose-driven, affordable innovation for the world.

Startup ecosystem and government support

There are two government schemes: the Design Linked Incentive (DLI) and the Production Linked Incentive (PLI). DLI focuses on semiconductor design, while PLI supports